

Review

The Role of AI in Academic Writing: Impacts on Writing Skills, Critical Thinking, and Integrity in Higher Education

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Abstract

Artificial Intelligence (AI) tools have transformed academic writing and literacy development in higher education. Students can now receive instant feedback on grammar, coherence, style, and argumentation using AI-powered writing assistants, like Grammarly, ChatGPT, and QuillBot. Moreover, these writing assistants can quickly produce completed essays and papers, leaving little else for the student to do aside from reading and perhaps editing the content. Many teachers are concerned that this erodes critical thinking skills and undermines ethical considerations since students are not performing the work themselves. This study addresses this concern by synthesizing and evaluating peer-reviewed literature on the effectiveness of AI in supporting writing pedagogy. Studies were selected based on their relevance and scholarly merit, following the Scale for the Assessment of Narrative Review Articles (SANRA) guidelines to ensure methodological rigor and quality. The findings reveal that although AI tools can be detrimental to the development of writing skills, they can foster self-directed learning and improvement when carefully integrated into coursework. They can facilitate enhanced writing fluency, offer personalized tutoring, and reduce the cognitive load of drafting and revising. This study also compares AI-assisted and traditional writing approaches and discusses best practices for integrating AI tools into curricula while preserving academic integrity and creativity in student writing.

Keywords: artificial intelligence (AI); writing skills; critical thinking; integrity; higher education

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1. Introduction

In today's society, artificial intelligence (AI) tools have become a part of many people's everyday lives [1]. These tools are powerful applications designed to help generate, edit, and improve written content [2]. They use natural language processing (NLP) models, like Open AI's GPT or Google's BERT, to analyze and generate human-like text [3]. The applications "learn" from massive amounts of text data found on the internet and through other sources. They can understand context, syntax, and tone, enabling them to create coherent sentences and paragraphs in various styles [4]. Moreover, as these tools have advanced, they are increasingly becoming essential for content creation in multiple fields because they offer efficiency and scalability for businesses, educators and students, and other creative writing professionals [5].

AI writing tools have advanced to the point where they can perform many different tasks, from drafting blog posts and emails to creating marketing copies and even writing complicated research papers [6,7]. What is impressive is that they can quickly produce

polished, grammatically correct, stylish, and engaging content in a fraction of the time it would take a human. Some advanced AI tools can even conduct sentiment analysis and tailor content to specific audiences and/or industries [6]. With additional machine learning algorithms, these tools can also adapt over time and learn from user feedback to improve their output. This level of adaptability makes them an invaluable resource for individuals and organizations looking to simplify and enhance their workflows [6].

It is therefore not surprising that many industries have adopted these tools. Fields as diverse as education, journalism, marketing, customer service, and blog writing have all adopted AI to some degree. In education, students are turning to these tools to help them draft essays or write papers [6,8]. Professionals in various industries use them to prepare reports, and non-native speakers find them invaluable for refining their language skills [6,9]. Many businesses are using these tools to automate routine communication tasks, including generating search-engine-optimized content and creating customer responses [6,10,11]. Moreover, these tools allow companies and individuals to enhance creativity, brainstorm ideas, and test alternative phrasing [12,13]. Peer-reviewed research highlights their utility for these purposes and in improving productivity. However, concerns remain about originality, authenticity, and whether generating content via AI might be undermining educational outcomes in courses related to writing development [12,13].

1.1. The Role of AI in Writing Development

Natural language processing (NLP) and machine learning algorithms enable AI to analyze text, detect errors, and provide feedback tailored to a writer's needs [12,14,15]. Some AI applications, like Grammarly and Hemingway Editor, focus more on grammar and writing mechanics, offering tools to help writers refine their sentence structure and improve the readability of their text [14,16]. Other tools, such as ChatGPT, go beyond surface-level corrections by assisting writers in generating ideas, writing original content, rephrasing sentences, and enhancing argumentation [9,13]. Additionally, reference management tools, such as Zotero and Mendeley, help writers organize references and ensure academic integrity. Since their emergence, these AI-driven tools have undergone significant advancements, becoming increasingly sophisticated. As a result, they are now increasingly regarded as critical parts of the writing process, particularly for writers who may face challenges with the linguistic and structural aspects of writing [17,18].

As these tools have become more available and accessible, they have significantly changed how students approach writing assignments [12]. Rather than relying solely on instructor feedback, students now have real-time access to automated applications that can help them refine their work. As a result, educational professionals have become increasingly concerned about whether these tools genuinely enhance writing skills and promote learning or simply foster dependency [12,19]. Moreover, educators are concerned about the impact of AI-powered writing tools on critical thinking skills and originality. Addressing these concerns is crucial to help educators integrate AI into their coursework effectively, ensuring that students develop their skills rather than rely on a tool as a long-term crutch. Furthermore, it is vital to understand the impact of AI on academic integrity [12].

1.2. Potential Benefits of AI in Writing Instruction

AI-powered writing tools offer numerous advantages, particularly for students in higher education [20,21]. For example, non-native English speakers can use these tools to enhance their proficiency in English writing [20,22]. Similarly, students with learning disabilities can use AI for clearer explanations, real-time corrections, and writing suggestions, thereby improving their overall writing proficiency [23]. Moreover, AI enables students to

engage with writing at their own pace, eliminating the need to wait for instructor feedback and thereby fostering self-directed learning [20]. In addition, by assisting students in drafting and revising complex academic texts, AI tools can reduce their cognitive load, enabling them to focus on higher-order thinking and critical analysis [24,25].

From an instructional perspective, AI tools can help educators assess student writing by identifying common errors and producing personalized instruction [2]. For example, certain AI programs are capable of delivering formative feedback, highlighting areas requiring improvement, and recommending revisions aligned with established writing conventions and best practices [13,20]. Moreover, in large classroom settings, where individualized instructor feedback may be limited, AI can serve as a valuable supplementary resource, thereby enhancing overall student learning outcomes [24].

1.3. Challenges and Ethical Considerations

While AI offers significant benefits in writing instruction, it is not without its challenges [13]. A primary concern among educators is the potential for students' over-reliance on AI-generated suggestions, which may lead to factual inaccuracies, diminished critical thinking, and a decline in independent problem-solving abilities [15]. Students who become too dependent on AI tools for grammar correction, paraphrasing, or content generation may struggle to develop essential writing competencies on their own [12]. Additionally, there are ethical concerns about AI-generated content, including issues of academic integrity, plagiarism, and originality [13,15]. Some AI tools obscure the distinction between acceptable paraphrasing and unethical text manipulation, making it more challenging for educators to assess the authenticity of student work [12].

Furthermore, while AI-generated feedback can be beneficial, it often lacks contextual accuracy and pedagogical appropriateness [8]. This is because AI tools cannot fully understand specific nuances of argumentation, tone, and discipline-specific writing conventions. This means that students may receive misleading, overly generic, or even factually incorrect feedback, which may potentially hinder their ability to meet academic expectations [12]. Moreover, a pressing question arises: if AI is conducting the research and writing, will students be able to develop necessary analytical and writing skills independently? Although the answer remains uncertain, this is a critical issue that warrants further investigation [12,23].

1.4. Purpose and Scope of the Review

Specifically, this paper will explore the following topics: (1) the impact of AI writing tools on students' ability to develop independent writing skills and critical thinking; (2) the benefits and limitations of AI-assisted writing tools in improving grammar, coherence, and argumentation; (3) the perceptions of students and educators regarding the effectiveness of AI writing tools in higher education; (4) the ethical concerns raised by AI-assisted writing, including issues related to plagiarism, originality, and academic integrity; (5) the comparison between AI-based writing tools and traditional instructional methods in terms of learning outcomes and skill improvement; and (6) the best practices for integrating AI tools into writing curricula to optimize student learning outcomes.

2. Method

We conducted a narrative review by systematically searching three academic databases, EBSCOhost, ERIC, and JSTOR, to identify relevant studies on the role of AI tools in academic writing. Additionally, we used Google Scholar as a supplementary resource to broaden our scope and identify relevant sources beyond specific databases. This narrative review aimed to synthesize the existing literature on AI-assisted writing, its effectiveness in educational

settings, and its implications for higher education. A narrative review is particularly suitable for exploring various aspects of a research topic, as it facilitates the integration of broad themes and concepts derived from multiple sources [26]. This approach allows for a comprehensive synthesis of complex or extensive research evidence, providing detailed descriptions and interpretations [26].

2.1. Search Tactic

Table 1 presents the search strategy used in this review. Boolean operators were applied to refine the search process, ensuring a precise and accurate identification of the relevant literature.

Table 1. Search strategy using Boolean operators.

Keywords	Boolean Operators
AI writing, artificial intelligence in writing	("AI writing" OR "Artificial intelligence in writing") AND ("academic writing" OR "higher education")
Natural language processing, AI text generation	("Natural language processing" OR "AI text generation") AND ("academic writing" OR "text improvement")
Grammar correction, automated proofreading	("Grammar correction" OR "Automated proofreading") AND ("academic writing" OR "university assignments")
AI-generated text structure, writing organization	("AI-generated text structure" OR "Writing organization") AND ("higher education" OR "research writing")
Plagiarism detection, AI ethics in writing	("Plagiarism detection" OR "AI ethics in writing") AND ("academic integrity" OR "university policies")
AI vs. human feedback, writing pedagogy	("AI vs. human feedback" OR "Writing pedagogy") AND ("academic writing" OR "university courses")
Student perception, educator perspective on AI writing	("Student perception" OR "Educator perspective on AI writing") AND ("higher education" OR "academic integrity")
AI in research writing, AI in literature review	("AI in research writing" OR "AI in literature review") AND ("scientific writing" OR "research methodology")
AI feedback, writing revision	("AI feedback" OR "Writing revision") AND ("editing process" OR "academic improvement")
AI in education, AI writing tools	("AI in education" OR "AI writing tools") AND ("writing skill development" OR "higher education")
Future of AI in writing, AI's impact on academic writing	("Future of AI in writing" OR "AI's impact on academic writing") AND ("technology in education" OR "writing trends")

2.2. Inclusion and Exclusion Criteria

The selection of studies for this review was guided by clearly defined inclusion and exclusion criteria, as presented in Table 2. This systematic approach ensured that this research included only the most pertinent and up-to-date articles aligned with the study's objectives.

Table 2. Inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
Publication Date	Studies published between 2023 and 2025, ensuring the most recent AI advancements in academic writing are included.	Studies published before 2023, as older research, may not reflect the latest developments in AI-assisted writing.
Study Type	Peer-reviewed journal articles, conference proceedings, systematic reviews, and opinion or commentary articles that critically examine AI's impact on academic writing.	Non-peer-reviewed sources, such as blog posts, news articles, and non-academic reports, as they lack scientific rigor.

Table 2. *Cont.*

Criteria	Inclusion	Exclusion
Language	English-language publications, ensuring accessibility and consistency in interpretation.	Non-English studies, unless a peer-reviewed translation is available, avoiding misinterpretation.
Relevance to AI in Academic Writing	Studies that focus on AI-powered writing tools, automated feedback, AI in research writing, and AI's role in higher education.	Studies focusing on AI in journalism, creative writing, marketing content, or corporate communications, which do not align with academic writing research.
Context and Population	Research on higher education students, university faculty, and academic researchers using AI for writing.	Studies focused on K–12 students, industry professionals, and corporate AI writing tools, as they do not align with the review's scope.
Methodological Rigor	Empirical studies, systematic reviews, comparative studies, and experimental research with transparent methodologies.	Opinion-based articles, informal case studies, or theoretical discussions without empirical validation.
AI Tools Analyzed	Studies evaluating AI tools, such as ChatGPT, Grammarly, and QuillBot, and AI-powered citation managers (e.g., Zotero and Mendeley) in academic writing.	Studies on traditional writing tools that do not use AI, such as manual proofreading software or simple grammar checkers.
Ethical and Pedagogical Considerations	Studies addressing AI ethics, academic integrity, plagiarism concerns, and AI's pedagogical role in education.	Studies that do not discuss AI ethics or its role in education, focusing solely on technical aspects.

2.3. Review Strategy

The study selection process began with an initial screening of articles by evaluating abstracts, introductions, and conclusions to determine their relevance and adherence to the inclusion and exclusion criteria. Articles that met the initial requirements were then qualitatively assessed in detail, where their content was analyzed more comprehensively. During this stage, duplicate studies were identified and removed to prevent redundancy.

To ensure the methodological rigor of the review, the study selection process followed the SANRA (Scale for the Assessment of Narrative Review Articles) guidelines [27], ensuring that only high-quality studies were included. After applying the inclusion and exclusion criteria and eliminating duplicates, 20 studies were finalized for inclusion in the review.

3. Results

Screening Outcomes

Out of the 261 articles initially retrieved, only 20 studies satisfied the eligibility criteria and were included in the final analysis. The 20 chosen studies employed various research methods, including systematic reviews, quantitative analyses, qualitative investigations, and mixed-method research. Table 3 presents a comprehensive summary of each study, including the Study Number, In-text Citation, Research Region, Population, Purpose of Study, Method, and Key Insights. Table 4 provides the core thematic areas examined in this review in regard to their significance in AI-assisted academic writing.

Table 3. Summary of data extracted from 20 studies.

Study No.	Citation	Study Region	Population	Purpose of Study	Method	Key Insights
S1	Gasaymeh et al., 2024 [12]	Jordan	University Students	To examine university students' familiarity, concerns, and perceived benefits of generative AI writing tools in academic work.	Quantitative	Students had moderate familiarity with AI tools, recognized benefits, and had concerns about misinformation and data security.
S2	Schei et al., 2024 [23]	Norway	University Students	To analyze students' perceptions and use of AI chatbots in higher education through a scoping review of empirical studies.	Review	Students perceive AI chatbots as useful for task assistance and learning but express concerns about accuracy, reliability, and potential negative impacts on critical thinking and creativity.
S3	Dwivedi et al., 2023 [6]	Cross-Country	Various Academic Studies	To explore the opportunities, challenges, and implications of generative AI (ChatGPT) in research, practice, and policy across different fields.	Commentary Article	ChatGPT presents both opportunities and challenges across education, business, and society, with debates on its ethical, legal, and practical implications.
S4	Marzuki et al., 2023 [2]	Indonesia	University Students	To examine the impact of AI writing tools on the content and organization of student writing, as perceived by EFL teachers.	Qualitative	AI writing tools improved students' writing quality, especially in content and organization, though concerns exist regarding over-reliance on these tools.
S5	Wang & Ren, 2024 [9]	Hong Kong	University Students	To examine how university students use generative AI tools for digital academic writing through a collaborative Wikibook project.	Mixed Methods	Students effectively used AI tools for structuring discourse and enhancing writing but expressed concerns about ethical use, critical thinking, and plagiarism.
S6	Khalifa & Albadawy, 2024 [13]	Cross-Country	Various Academic Studies	To explore how artificial intelligence enhances academic writing and research across various domains.	Review	AI significantly supports academic writing in idea generation, literature synthesis, editing, and publishing, though ethical considerations remain crucial.
S7	Zhai et al., 2024 [15]	Cross-Country	Various Academic Studies	To investigate the contributing factors and effects of over-reliance on AI dialogue systems in research and education, particularly their impact on students' cognitive abilities.	Review	Over-reliance on AI weakens decision-making, critical thinking, and analytical skills, with ethical concerns like biases and AI hallucinations.
S8	Fiorillo, 2024 [7]	Cross-Country	Content Analysis	To reevaluate the role of AI writing tools in scientific integrity, addressing concerns about their ethical use and potential benefits.	Commentary Article	AI writing tools can enhance clarity and efficiency in research but require ethical oversight to prevent biases and misinformation.
S9	Gawlik-Kobylińska, 2024 [19]	Poland	University Students	To explore students' perspectives on integrating AI into scientific collaboration, particularly in academic writing and creating scientific posters.	Qualitative	AI enhances efficiency and idea generation in academic collaboration but raises concerns about technical difficulties, over-reliance, and ethical considerations.
S10	Kim et al., 2025 [28]	China	University Students	To explore students' perceptions and experiences with generative AI-assisted academic writing.	Qualitative	Students viewed AI as a multi-tasking assistant, virtual tutor, and digital peer in writing. AI improved writing efficiency and organization but raised concerns about accuracy, biases, and ethical use.
S11	Artyukhov et al., 2024 [29]	Cross-Country	Academic Publications	To analyze the relationship between Sustainable Development Goal 4 (SDG 4), academic integrity, and AI in education, determining whether AI poses a challenge or an opportunity for academic integrity.	Qualitative	AI can enhance personalized learning and educational accessibility but also poses risks to academic integrity, such as plagiarism and AI-generated content misuse.

Table 3. Cont.

Study No.	Citation	Study Region	Population	Purpose of Study	Method	Key Insights
S12	Abani et al., 2023 [16]	Cross-Country	Content Analysis	To discuss the potential benefits and risks of ChatGPT in scientific writing, particularly in veterinary neurology.	Commentary Article	ChatGPT enhances accessibility and efficiency in research but raises concerns about AI-generated plagiarism, misinformation, and ethical use in academic publishing.
S13	McIntire et al., 2024 [30]	North America	Content Analysis	To reframe academic integrity by shifting the discussion from ethics to pragmatism in addressing plagiarism and cheating.	Qualitative	Academic dishonesty is seen as a pragmatic choice. Universities should highlight its impact on learning rather than just ethics.
S14	Rumanovská et al., 2024 [31]	Slovakia	University Students	To assess the effectiveness of education in reducing plagiarism among university students and explore factors contributing to plagiarism.	Quantitative	Education helps reduce plagiarism but alone is insufficient. Additional preventive and repressive measures are needed, such as stricter academic policies and better monitoring.
S15	Ruiz-Rojas et al., 2024 [21]	Cross-Country	University Students	To assess the impact of generative AI tools on students' critical thinking and collaboration in higher education.	Mixed Methods	AI tools enhance critical thinking and collaboration, but students require continuous training and technical support for effective use.
S16	Ahn, 2024 [24]	South Korea	Lifelong Learners (Adults in Higher Education and Professional Development)	To examine the impact of AI-powered e-learning on lifelong learners' performance and knowledge application.	Quantitative	AI tools improve learning efficiency, job performance, and knowledge application, but usability and confidence in AI influence adoption.
S17	Chen & Gong, 2025 [20]	China	University Students	To examine the role of AI-assisted learning in academic writing for Chinese as a Second Language (CSL) students.	Mixed Methods	AI-assisted learning improves writing outcomes and motivation but raises concerns about over-reliance, ethical issues, and the reliability of AI-generated content.
S18	Dhanapal et al., 2024 [32]	Saudi Arabia	University Students	To compare the effectiveness of AI-based instruction with traditional teaching methods in language learning.	Mixed Methods	AI-based instruction enhances language learning outcomes by providing personalized feedback, motivation, and engagement, but effectiveness varies among learners.
S19	Lin et al., 2023 [33]	Cross-Country	Academic Publications	To examine how AI-driven intelligent tutoring systems (ITSs) contribute to sustainable education.	Review	AI-powered ITSs enhance personalized learning and student engagement, but challenges like biases, transparency, and digital accessibility remain.
S20	Kamalov et al., 2023 [5]	Cross-Country	Academic Publications	To review AI's impact on education, analyzing its applications, benefits, and challenges.	Review	AI enhances personalized learning, intelligent tutoring, and assessment automation but raises ethical concerns about biases, data privacy, and academic integrity.

Table 4. Summary of key themes in AI-assisted writing tools in education.

Core Theme	Thematic Subdivisions	Significance and Application
Enhancing Writing Skills and Efficiency	Grammar and syntax correction; Sentence structure improvement; Writing clarity and coherence.	AI tools refine language proficiency by identifying and correcting grammatical errors, enhancing structure, and ensuring clarity.
Support for Logical Argumentation	Providing structured content suggestions; Generating counterarguments; Strengthening persuasive and evidence-based writing.	AI tools aid in argument construction by offering logical structuring, providing counterarguments, and enhancing coherence in academic writing.

Table 4. Cont.

Core Theme	Thematic Subdivisions	Significance and Application
Improving Accessibility for Non-Native Speakers	Enhancing vocabulary and phrase selection; Providing sentence reconstruction assistance; Offering better contextual and idiomatic suggestions.	AI-powered suggestions improve English proficiency by refining vocabulary, restructuring complex sentences, and making idiomatic expressions more accurate.
Challenges of AI-Assisted Writing	Over-reliance on AI for content generation; Reduced independent critical thinking; Ethical concerns and plagiarism risks.	Excessive reliance on AI may lead to diminished critical thinking skills, dependency on automated writing, and concerns regarding academic integrity.
Academic Integrity Concerns	AI-generated content as potential plagiarism; Lack of proper source attribution; Questions about authenticity and authorship.	AI-generated text blurs traditional plagiarism definitions, requiring clearer policies on appropriate AI usage in academia.
Ethical Considerations in AI Writing	Establishing fair usage policies; Increasing AI transparency requirements; Developing institutional guidelines on AI-assisted writing.	Institutions need clear guidelines to define the ethical boundaries of AI-generated content while preserving student originality.
Limitations in Understanding Context and Nuance	Misinterpretation of meaning and intent; Poor handling of sarcasm, irony, and tone shifts; Difficulty processing abstract and creative writing.	AI tools struggle with context-sensitive writing, tone shifts, and complex abstract reasoning, necessitating human oversight.
Balancing AI Use and Independent Skill Development	AI as a complementary tool rather than a substitute; Encouraging critical engagement with AI-generated content; Instructor-guided integration of AI in writing assignments.	AI should support, rather than replace, independent thinking by being incorporated as an aid in the writing process under proper educational guidance.
Perceptions of Students on AI Writing Tools	Increased confidence in writing ability; Faster drafting and editing process; Concerns about originality and reliance on AI.	Students view AI as a useful tool for efficiency and feedback, but some worry about diminished deep learning and authentic skill development.
Perceptions of Educators on AI Writing Tools	Support for structured writing and organization; Ethical concerns regarding over-reliance on AI; Need for well-defined institutional policies on AI use.	Educators recognize AI's benefits but stress the importance of maintaining academic integrity and teaching students responsible AI use.
AI vs. Traditional Writing Instruction	AI enhances efficiency and speed; Traditional instruction fosters more profound learning and long-term skill retention; AI is best used as a supplementary learning tool.	AI tools improve writing efficiency but lack the depth of traditional instruction, making a balanced approach necessary for effective writing development.
Best Practices for AI Integration in Writing Curricula	AI as a writing aid, not a generator; Encouraging active student engagement; Developing ethical AI usage guidelines.	AI should be framed as a tool to refine and support writing skills while ensuring students remain the primary authors of their work.
Encouraging Critical Thinking and Argumentation	Assignments requiring AI analysis and critique; Student-centered editing and revision with AI; AI-prompted idea generation to support creativity.	Educators can design tasks where students critically evaluate AI-generated content, ensuring active engagement in the learning process.

4. Discussion

4.1. The Benefits of Using AI-Assisted Writing Tools

Since students have quickly adopted AI writing tools, like ChatGPT, Grammarly, and QuillBot, to complete their assignments, the question of how this affects their ability to develop independent writing skills and critical thinking inevitably arises [12,23]. These tools provide invaluable support by offering instant grammar corrections and content suggestions. They can even generate entire essays and help students formulate arguments, saving the students significant time and improving their language use, accuracy, and coherence, among other benefits [12].

4.1.1. Improving Grammar and Writing Efficiency

AI tools offer several advantages in terms of improving grammar, coherence, and arguments. They can improve the text's clarity, enhance the language, and help structure logical arguments [6,12]. AI writing tools, such as Grammarly, enable students to improve their grammar and language accuracy. Many not only detect errors but also provide explanations, so that students can actively work to improve their writing [2]. When utilized as intended, students are likely to refine their language skills over time, thereby reducing the occurrence of grammatical mistakes [9]. Moreover, these tools can help non-native English speakers improve their language proficiency by offering suggestions for sentence construction and proper word usage. The instant nature of the feedback provided also reduces the need for external proofreading, thereby improving efficiency [9].

One of the most significant benefits of AI writing tools is that they can enhance writing efficiency [13,15]. Through real-time grammar corrections, structural improvements, and suggestions for more coherent writing, these tools enable students to produce more polished work with less effort [13]. Moreover, AI-powered platforms, such as Grammarly, not only point out mistakes but also explain the grammatical rules, allowing students to learn from their errors and improve their writing [2]. Additionally, AI tools that make content suggestions can help students overcome writer's block by providing prompts, sentence suggestions, and guidance on structuring an essay. These features are undoubtedly beneficial for students who struggle with organizing their thoughts and ideas in their writing [2].

4.1.2. Strengthen Logical Reasoning, Critical Thinking, and Evidence-Based Writing

Beyond correcting grammatical errors, AI writing tools can improve the overall coherence of writing by offering suggestions to enhance logical flow and the organization of concepts within the text [7]. They achieve this by analyzing sentence structures, transitions, and paragraph organization to generate suggestions for making the content more readable and engaging. This process helps eliminate redundancy, improve word choice, and enhance clarity, enabling writers to produce clear, concise, and well-organized content [2].

While improving the 'nuts and bolts' of a student's writing is valuable, AI tools can also help students strengthen their logical reasoning and evidence-based writing [2]. These tools can assist students in identifying credible sources, improving transitions between ideas, and ensuring overall consistency in arguments [2]. Some tools, like ChatGPT, can also generate counterarguments, suggest rebuttals, and refine persuasive writing. These functionalities can help students stay engaged with the material and improve their ability to construct well-reasoned arguments [2].

4.2. The Limits and Challenges of AI-Assisted Writing Tools

While AI-assisted writing tools offer numerous benefits, there are growing concerns about their potential drawbacks [19]. Some key questions arise: How accurate is AI-generated content? Are students truly engaging in the writing process, or are they merely depending on this technology to fulfill course requirements? What about originality and authenticity?

4.2.1. The Problem of Over-Reliance

One of the biggest concerns is that students are becoming over-reliant on AI tools, which could cause them to become dependent on them rather than develop their own skills [19]. Students no longer have to construct their arguments or structure their essays to explain their points of view; instead, they simply let the AI writing tool perform the work. This can ultimately diminish their creativity and critical thinking abilities [2]. Writing is not

only about assembling words into sentences that make sense; it is also about engaging with the material in a way that encourages students to analyze information critically and apply independent reasoning to the subject matter [28]. While AI can provide well-structured content, it does little to encourage the cognitive effort necessary for students to form their own ideas and arguments. This lack of engagement can easily lead to a decline in students' critical thinking skills [2,15,28].

Students who rely too much on AI-generated text do not fully engage with the complexity of their subject matter. As a result, they miss opportunities to develop their reasoning [15]. Part of the purpose of a writing assignment is to challenge students to analyze, synthesize, and present their original ideas in their own words [28]. If students rely on AI to complete tasks with minimal effort, it undermines the purpose of education and hinders instructors' ability to accurately assess their skills [15,29]. Additionally, there are several concerns about plagiarism and the authenticity of AI writing. Most AI tools utilize information from internet sources but often fail to verify factual accuracy or properly credit sources [28].

4.2.2. Ethical Concerns

As AI-generated content advances, educators and academic institutions must promote ethical writing practices while responsibly incorporating AI [7,13]. A key ethical issue tied to AI-assisted writing is plagiarism. Traditionally, plagiarism involves presenting someone else's work as one's own without proper attribution [30,31]. AI blurs this definition, as its generated content, while not directly copied from existing sources, is still created by an external tool rather than a student's independent intellectual effort [15,30]. Many students may unknowingly engage in academic dishonesty by submitting AI-generated or paraphrased content without realizing its ethical implications [6,9].

Moreover, AI-powered paraphrasing tools, like QuillBot, enable users to rephrase extensive passages with minimal effort, often producing surface-level changes that lack meaningful engagement with the original ideas. Such practices could still be considered plagiarism, creating new challenges for educators trying to uphold ethical standards [6,9]. The growing use of AI in academia complicates the detection and definition of plagiarism, prompting institutions to reconsider how they enforce academic integrity in this evolving digital landscape [15,30].

Originality is another major ethical issue for AI-assisted writing [6,12,15]. Writing is a fundamental academic exercise designed to encourage independent thinking, creativity, and the development of original arguments [6,9]. However, when students rely on AI-generated content, they may bypass the critical thinking and intellectual effort required to develop their ideas [12,15]. AI writing tools provide polished, well-structured responses but often lack genuine insight, critical analysis, and the writer's unique voice. This excessive reliance on AI may weaken students' ability to engage deeply with their subjects and hinder their capacity to produce original work [6,9].

Furthermore, some AI-generated content may be based on pre-existing data and training sets that include inaccuracies or biased perspectives, leading to concerns that AI models could inadvertently produce responses that resemble previously published or even discredited materials. This raises questions about the authenticity and reliability of AI-generated work and its role in higher education [19,28].

Academic integrity is another key ethical concern when discussing AI-assisted writing [19,28]. Academic institutions emphasize the importance of honesty, effort, and the demonstration of independent learning in written assignments. When students use AI tools to generate entire essays, they bypass the learning process and fail to develop essential writing and reasoning skills [6,12,15]. While AI tools can assist with grammar, structure, and

clarity, their misuse poses a serious threat to academic integrity policies [19,28]. Universities now face the challenge of defining acceptable AI use and distinguishing it from unethical practices. Some institutions have already updated their academic integrity policies to address AI-related concerns, introducing AI-detection software and stricter guidelines on AI-assisted writing [6,19].

4.2.3. Context and Nuances

Finally, another limitation of AI writing tools is their inability to fully understand context and nuanced language [2,28]. While AI is excellent at detecting common grammatical errors and structural issues, it sometimes fails to grasp the intended meaning of a sentence. AI-generated suggestions may be grammatically correct but contextually inaccurate [28].

AI writing tools often struggle with context and nuanced language, particularly when handling idiomatic expressions or culturally specific phrases [2,28]. Similarly, AI tools often fail to grasp subtle tonal shifts, such as sarcasm or irony. These limitations highlight the need for human involvement to ensure contextual accuracy and a deeper understanding [19,28]. Additionally, AI struggles with creative and subjective writing, such as poetry or literary analysis, which require a deeper comprehension of emotions and themes. This limitation further highlights the importance of human judgment in refining AI-assisted writing [19,28].

4.3. Addressing the Challenges of Using AI-Assisted Writing Tools

Addressing these challenges requires striking a balance between using AI tools and fostering students' independent writing skills [2,28]. Educators must recognize that students will likely use AI for writing assistance and thus must develop strategies to integrate these tools effectively into learning [2,9]. This integration ensures that students actively engage with the content, think critically, and employ creativity and reasoning in their work.

One practical approach is to position AI-generated content as a starting point rather than a final product [28]. For instance, instructors can ask students to critique, revise, or expand upon AI-generated text by identifying weaknesses, questioning assumptions, and improving clarity. This process encourages close reading, critical thinking, and the development of well-reasoned arguments, allowing students to add their own perspectives [6].

Another strategy is to assign collaborative tasks in which students evaluate, compare, and integrate AI-generated suggestions with their own ideas [21,28]. These activities deepen their understanding of the topic, enhance creativity, and build teamwork skills [21]. Such practices align with constructivist learning theories, which emphasize active participation and reflection as vital components of meaningful learning [34]. By incorporating these methods, educators can guide students in using AI tools responsibly while actively engaging in their learning journey, fostering both technological literacy and independent intellectual growth [21].

In summary, AI-assisted writing tools can substantially improve grammar, coherence, and argumentation, making writing more accessible and efficient [2,28]. These tools help users correct errors, enhance clarity, structure logical arguments, and improve communication skills [7,13]. However, they also have limitations, including over-reliance, ethical concerns, and contextual misunderstandings [6,28]. To maximize their benefits while mitigating drawbacks, students and professionals should use AI writing tools as supplementary aids rather than replacements for critical thinking and independent writing [7,13,16]. By striking a balance between AI assistance and human creativity, writers can harness the power of technology while preserving their ability to think critically and express themselves effectively [7,13,16].

4.4. The Perceptions of Students and Educators Regarding the Effectiveness of AI Writing Tools in Higher Education

Understanding how students and educators perceive AI writing tools is essential [2,12]. These tools have transformed the way students approach writing assignments. While many see them as valuable assets, others have expressed concerns. The perspectives of both students and educators provide valuable insights into the advantages and challenges of AI-assisted writing in academia [12,13,24].

4.4.1. The Students' Perspectives

Naturally, many students view AI writing tools as beneficial aids that help them improve the quality and efficiency of their writing [13]. Students appreciate the instant feedback on grammar, syntax, and coherence, enabling them to refine their work without waiting for instructor feedback [12]. AI writing tools are especially valuable for non-native English speakers. These tools help bridge language barriers by offering alternative word choices and precise grammatical corrections [9,13]. Moreover, they help students structure their ideas in clear, concise, and engaging ways that enhance the overall readability of the content. Undoubtedly, accessibility and convenience are aspects most students appreciate [9,13].

Not all students see AI writing tools as purely beneficial [12]. Many have expressed concerns that these tools can undermine critical thinking, logical reasoning, and creativity [21,23]. Many students want to engage deeply with the content and improve their writing skills. Students have also expressed concerns about AI tools failing to capture the complexity of specific arguments or the nuances present in academic writing. These students want to engage intellectually with the material to analyze it critically and produce a well-reasoned argument [12,20,23]. They see AI as a tool to assist in this process, but they do not want to be overly reliant on it to complete their academic tasks. They want to develop their voice and argumentative skills to foster the professional growth they need to succeed [12,21,23].

4.4.2. Educators' Perspectives

While many educators acknowledge the clear advantages of AI in helping students improve their grammar and content organization, they have mixed opinions on the effectiveness of using these tools in academic settings [23]. Some professors encourage the use of AI-assisted tools to supplement student assignments. They believe AI can help facilitate the writing process while promoting the essential cognitive skills required for the material. These tools are also helpful to instructors in reducing the burden of repetitive feedback to correct grammar errors and improve coherence [9]. Thus, they can free educators to focus more on the content and argumentation.

However, many educators are also concerned about how these tools affect academic integrity and learning outcomes [6,12]. They have serious ethical concerns about the originality of AI-generated content, and they fear it encourages a passive approach to learning, with the students merely relying on technology to complete their assignments. This further leads to challenges in assessing students' writing abilities and critical thinking skills [9]. In response to their concerns, some higher education institutions have begun to implement guidelines for the ethical use of AI. For example, they may require students to disclose their use of these writing tools. Instructors, meanwhile, have integrated guidance into their classroom assignments on using AI responsibly to maintain academic integrity [9]. They also encourage students to critically evaluate AI-generated content and use their reasoning and creativity to expand on the writing by adding their own perspectives. While

this does not solve all problems, it can help mitigate the most significant issues of over-reliance and diminished critical thinking skills [21].

4.5. A Comparison Between AI-Based Writing Tools and Traditional Instruction Methods in Terms of Learning Outcomes and Skill Retention

The debate over the effectiveness of AI writing tools versus traditional instruction centers on two critical aspects: learning outcomes and instructional effectiveness [32]. While AI tools provide immediate and data-driven assistance, traditional methods focus on in-depth comprehension, long-term retention, and cognitive engagement. A comparative analysis of these two approaches is essential to fully understand the similarities and differences [32].

4.5.1. Impact on Learning Outcomes

A significant difference between AI-based writing tools and traditional instruction is their impact on learning outcomes [32]. AI tools improve writing efficiency by providing real-time corrections for grammar, sentence structure, and word choice. Students can produce polished texts quickly, which helps reduce their frustration and makes writing more accessible, particularly for non-native English speakers [20]. AI can also help with brainstorming to assist students in generating ideas and logically structuring their essays. In contrast, traditional instruction methods foster deeper learning by requiring students to actively engage in the writing process [9,21] (Ruiz-Rojas et al., 2024; Wang & Ren, 2024). Teachers provide students with personalized feedback and guide them through argument development, critical thinking, and analytical writing. While AI primarily focuses on surface-level improvements, traditional instruction enables students to internalize writing principles and develop a deeper understanding of the reasoning behind grammatical and structural rules [9,32].

While AI tools may be more efficient, traditional instruction significantly contributes to skill retention [32]. Writing is more than producing grammatically correct, coherently structured sentences; it requires developing cognitive abilities, such as critical thinking, argument construction, and independent reasoning. Traditional instruction encourages students to practice these skills over time, reinforcing long-term retention by incorporating exercises, discussions, and feedback loops [21]. AI provides students with a polished product, but it does not require them to engage fully in the writing and learning process. Although AI enhances short-term writing proficiency, it does not foster long-term retention of writing skills to the same extent as traditional instruction [32].

4.5.2. The Role of Instructional Effectiveness

Another key distinction between AI-based writing tools and traditional instruction lies in the role of feedback within the learning process [12,32,33]. AI tools can provide automated, standardized feedback based on linguistic patterns and pre-set algorithms. While these tools provide instant and consistent feedback, they lack the depth and contextual understanding inherent in human assessment [32]. AI tools also struggle to assess argument strength, originality, and rhetorical effectiveness [15]. Traditional instruction, however, provides comprehensive and tailored feedback that addresses individual learning needs [32]. Effective instructors can identify students' weaknesses, offer detailed explanations, and foster improvements through various writing assignments. This personalized approach helps students refine their writing over time and foster a deeper understanding of the principles of strong writing [21].

Another key element for effective instruction is collaborative learning through peer reviews and discussions, which significantly contribute to skill development [21]. As students participate in classroom discussions, examine sample texts, and receive constructive

feedback from instructors and peers, their comprehension and critical thinking skills—especially in relation to their writing—significantly improve [20]. In contrast, AI operates in isolation, offering corrections but lacking the ability to facilitate dialogue or peer interactions. A critical aspect of writing lies in the social benefits gained through discussions, debates, and collaboration—elements that AI cannot fully replicate [32].

4.5.3. Accessibility and Scalability

Two areas where AI writing tools have a distinct advantage over traditional methods are accessibility and scalability [5]. AI tools can provide students with continuous, on-demand assistance, which traditional instruction cannot offer [28]. Students can access AI tools outside the classroom, allowing them to revise their work independently without waiting for instructor feedback. This capability is particularly beneficial for learners who lack access to personalized tutoring but still require additional writing support [20].

Moreover, AI tools are effective in facilitating large-scale writing improvement, particularly in educational institutions with high student-to-instructor ratios, where individualized feedback from instructors is challenging to provide. In comparison, traditional instruction relies heavily on teacher involvement, which can limit the ability to provide all students with immediate and individualized assistance [20].

4.6. Best Practices for Integrating AI Tools into Writing Curricula to Foster Student Learning

As discussed above, though AI can help students improve writing efficiency by offering continuous accessibility and real-time corrections, traditional instruction methods offer deeper learning, improved skill retention, and more comprehensive feedback [32]. AI writing tools serve as valuable supplements to help students improve their writing mechanics, though they cannot replace the cognitive engagement and critical thinking skills necessary for long-term writing proficiency. This suggests that the most effective approach for educators is to integrate AI as a supportive tool while preserving traditional instruction [32]. A balanced approach is likely the best solution for helping students develop strong writing skills. Instructors should emphasize the use of AI for immediate guidance, while designing assignments that promote reflective and analytical writing practices, fostering independent learning and lasting skill development [32]. Creating a balanced approach that integrates AI writing tools while ensuring the development of critical thinking and independent writing skills requires positioning these tools as complementary ones rather than as replacements for writing instruction [21].

4.6.1. AI as a Learning Aid

One way to achieve this is to use AI as a learning aid rather than a content generator [9]. Educators can implement this approach by designing structured assignments in which students first draft their content and subsequently use AI to identify areas for revision. This approach allows students to develop their creativity and gain a deeper understanding of writing mechanics while benefiting from AI-driven feedback. Through this process, students can improve sentence structure, refine word choices, and enhance coherence [9].

Another effective strategy is to have students critically evaluate AI-generated content. Instructors should emphasize that AI tools generate content based on algorithms, which often lack an understanding of the content's context, tone, or deep meaning. Instructors can foster analytical thinking skills by engaging students in analyzing AI-generated revisions for appropriateness [21]. Classroom discussions can then focus on distinguishing instances where AI-generated suggestions enhance clarity from those where they misinterpret the intended meaning. This approach encourages students to actively participate in their learning rather than passively accepting AI-generated corrections [21].

4.6.2. AI Collaboration

Another effective pedagogical approach is to integrate collaborative learning activities into the curriculum [21]. Having students work together to refine their writing and compare AI-generated revisions with peer feedback can expose them to diverse perspectives that contribute to improved writing [28]. Combining AI feedback with human feedback helps students recognize the importance of using both in a comprehensive learning process. It is also important to reinforce the importance of human judgment in writing [28]. Moreover, educators can incorporate exercises where students analyze AI-generated content to identify strengths and weaknesses, enhancing their critical thinking and analytical skills to further improve their writing [28].

4.6.3. Individualized Learning

Educators can also use AI to support personalized learning [5]. Given that AI tools can provide instant feedback, students can work at their own pace, and those requiring additional support can benefit from AI assistance without the necessity of one-on-one tutoring [33]. Educators can also use AI-generated reports on students' common writing errors to tailor their instruction and identify patterns they can address in targeted lessons. This approach enables instructors to utilize personalized AI-generated feedback without diminishing the role of traditional instruction [12,33].

4.6.4. Critical Thinking and Argumentation

To ensure that students can build and maintain a strong foundation in critical thinking and argumentation, educators must design assignments that require deeper engagement with writing [12,21]. For example, they can assign students the task of explaining their revisions after utilizing AI tools, thereby demonstrating their understanding of the changes. This approach compels students to reflect on their writing process and remain actively involved in their learning [21]. Educators can also use AI-generated prompts to stimulate student creativity, ensuring that students continue to develop original ideas and construct well-founded arguments [12,32].

4.6.5. Ethical Considerations

Finally, regarding the ethical use of AI writing tools, educators and institutions must establish clear guidelines specifying the appropriate and inappropriate use of AI in academic work [29]. Policies should clearly differentiate between the use of AI for grammar and mechanical support and the complete reliance on AI-generated text. Transparency in AI usage ensures ethical guidelines are followed [15]. There should be open discussions about the ethical implications of AI, and educators should leverage these discussions to help students develop academic integrity and avoid potential pitfalls associated with excessive dependence on AI-generated content [29]. These guidelines should be regularly reviewed and updated to reflect the evolving capabilities of AI and the changing landscape of academic integrity.

4.7. Limitations of the Current Review

This review employs a narrative methodology, which is well suited for synthesizing emerging evidence and identifying thematic insights across a developing field. At the same time, narrative reviews are inherently interpretive and less replicable than systematic or scoping reviews, which means that the findings should be understood as exploratory reflections rather than definitive tests of theory. Future research may build on this foundation by employing systematic or mixed-method approaches that enable more comprehensive theoretical testing. The breadth of the search strategy allowed for the inclusion of diverse

perspectives on AI in academic writing. However, the number of studies that met the inclusion criteria was limited, reflecting both the novelty of the field and the varied ways in which this topic is being investigated. While this constraint narrows the empirical base, it also highlights the need for further targeted studies and provides directions for future inquiry. Another consideration is the scope of the research questions. Addressing multiple dimensions of AI-assisted writing has enabled a broad view of its potential benefits and challenges. Yet, the questions were not all closely interrelated through a single theoretical lens, which may have reduced the depth of engagement with particular frameworks. At the same time, this breadth offers a useful starting point for identifying which aspects warrant deeper theoretical exploration in subsequent studies. The conclusions drawn from this review provide valuable insights into patterns and concerns surrounding AI in academic writing. These insights should be read as an informed synthesis of current knowledge rather than as conclusive generalizations, positioning this work as a platform for further empirical and theoretically grounded research.

4.8. Future Research Directions

To address the limitations discussed above, future research should emphasize longitudinal empirical studies that examine the impact of AI writing tools on learning outcomes and skill retention. Tracking students across multiple academic years would provide clearer insights into how these tools influence writing development, critical thinking, and overall academic performance. Comparative studies between students who use AI writing tools and those who rely on traditional instruction could also clarify the differential effects on writing proficiency and long-term retention. Broadening the scope of research beyond higher education is equally important. Examining how AI affects secondary students, non-native English speakers, and professionals in diverse fields would help educators and policymakers understand its influence across varied populations, enabling more targeted and inclusive integration strategies.

Attention should also be directed toward institutional policies and practices. Evaluating the effectiveness of regulations designed to ensure ethical AI usage and academic integrity would provide practical insights into governance and implementation. Evidence from such studies could guide institutions in adopting best practices that balance innovation with responsibility. Technological developments also present an important area of investigation. As AI tools evolve, research should consider improvements in contextual understanding, creativity support, and domain-specific applications. Exploring methods to reduce algorithmic biases and strengthen AI's ability to assess originality, argument quality, and rhetorical effectiveness will enhance their usability in educational settings.

Another essential avenue involves the psychological and behavioral dimensions of AI use. Studies on student motivation, engagement, and learning attitudes can reveal whether reliance on AI encourages active participation or fosters passivity. Insights from this line of inquiry will support the creation of guidelines that promote responsible use while safeguarding critical thinking and independent skill development. By addressing these areas, researchers and educators can develop a more balanced and evidence-based understanding of AI in writing instruction, ensuring that its integration supports both innovation and the fundamental goals of education.

5. Conclusions

AI-based writing tools are transforming writing instruction by providing real-time feedback, improving grammatical accuracy, and enhancing textual coherence. These tools can provide immediate and continuous corrections and assistance, offering a distinct advantage over traditional instruction methods, which are often limited by instructors'

availability and time constraints. As a result, they have made writing more efficient and accessible, particularly for non-native English speakers.

Despite these benefits, traditional instruction methods remain superior in fostering deep cognitive engagement, critical thinking, and long-term skill retention. In short, while AI tools enhance surface-level writing mechanics, they cannot replace the analytical depth and originality developed through traditional instruction and personalized feedback from educators. This underscores the value of a hybrid approach—using AI for mechanical improvements while relying on traditional methods to develop argumentation and analytical reasoning. Such an approach is likely to yield the best learning outcomes for students.

The ethical concerns surrounding AI-assisted writing primarily revolve around plagiarism, originality, and academic integrity. When students misuse AI-generated content, they bypass the intellectual effort essential for deep learning, which can stifle personal creativity and result in standardized, unoriginal submissions. Moreover, since AI models lack the ability to fully grasp nuanced writing, evaluating originality beyond basic structural improvements becomes challenging. This necessitates that educational institutions and instructors establish clear guidelines on AI usage to ensure students critically engage with both the subject matter and AI-generated suggestions. Implementing these guidelines can help curb over-reliance on AI and prevent ethical violations that arise from uncritically accepting AI-generated content.

Because AI is a relatively recent development, research on AI-assisted writing is still in its early stages. There is limited empirical data on the long-term effects of AI on student learning and cognitive development. Moreover, while AI tools are widely used in higher education, their impact on younger students, professional writers, and various academic disciplines remains largely underexplored. Additionally, improving AI's ability to effectively evaluate argument strengths, rhetorical effectiveness, and contextual intricacies represents areas ripe for future research. Other critical areas include how AI influences student motivation, engagement, and skill retention, as well as how AI-generated feedback compares to human instruction in developing writing proficiency over time.

A blended instructional approach using AI to enhance writing accuracy while employing traditional instruction to cultivate deeper-level learning appears to be the most effective strategy. This is important for helping students take ownership of the intellectual construction of their arguments. For example, exercises that require students to explain and justify the revisions suggested by AI tools are an effective way for instructors to ensure students actively engage in the learning process. Additionally, collaborative exercises like peer reviews can further help balance AI-assisted writing and traditional learning methods.

It is also crucial for educational institutions to implement policies that regulate AI usage to promote transparency, ethical use, and academic integrity. Higher education institutions and secondary schools should establish clear guidelines on AI use, ensuring that assessments focus on deeper learning outcomes, such as argument construction and defense. This will help prevent students from becoming overly dependent on AI-generated content. Moreover, instructors should introduce AI literacy programs to educate students on both the benefits and limitations of AI tools, as well as the importance of responsible, ethical usage in academic settings. While AI tools provide valuable support particularly in improving grammar and writing coherence, their integration into curricula must be carefully managed to preserve originality, critical thinking, and ethical academic practices. Traditional instruction remains indispensable in writing instruction, and AI should serve as a supplement rather than a replacement.

Future research should explore the long-term effects of AI on students' ability to write independently, creatively, and ethically. A well-balanced approach that strategi-

cally integrates AI while maintaining academic integrity will yield the most effective learning outcomes.

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